Design & Implementation of a Payroll Processing System

Problem Statement: A Company needs a simple Payroll Processing System to track its employees, their attendance, and their monthly salary payouts. The new database system will replace manual paperwork to prevent errors and ensure absolute data accuracy. The database must store and organize information across these six core business concepts:

● Departments: The company has several organizational units. Each department has a unique dept_code, a dept_name, and an annual_budget.

● Employees: Every employee has a unique emp_id, first_name, last_name, email, and belongs to exactly one department.

● Salary Tiers (Designations): To keep salaries fair, pay scales are fixed by job titles. Each tier has a unique desig_code, a job_title, and a fixed base_salary. An employee's profile links directly to one designation code, meaning their base salary is entirely determined by their job title.

● Attendance Logs: At the end of every month, HR records an employee's attendance variables. Each log entry tracks an employee for a specific cal_month and cal_year, recording the total days_present and any unpaid_leaves.

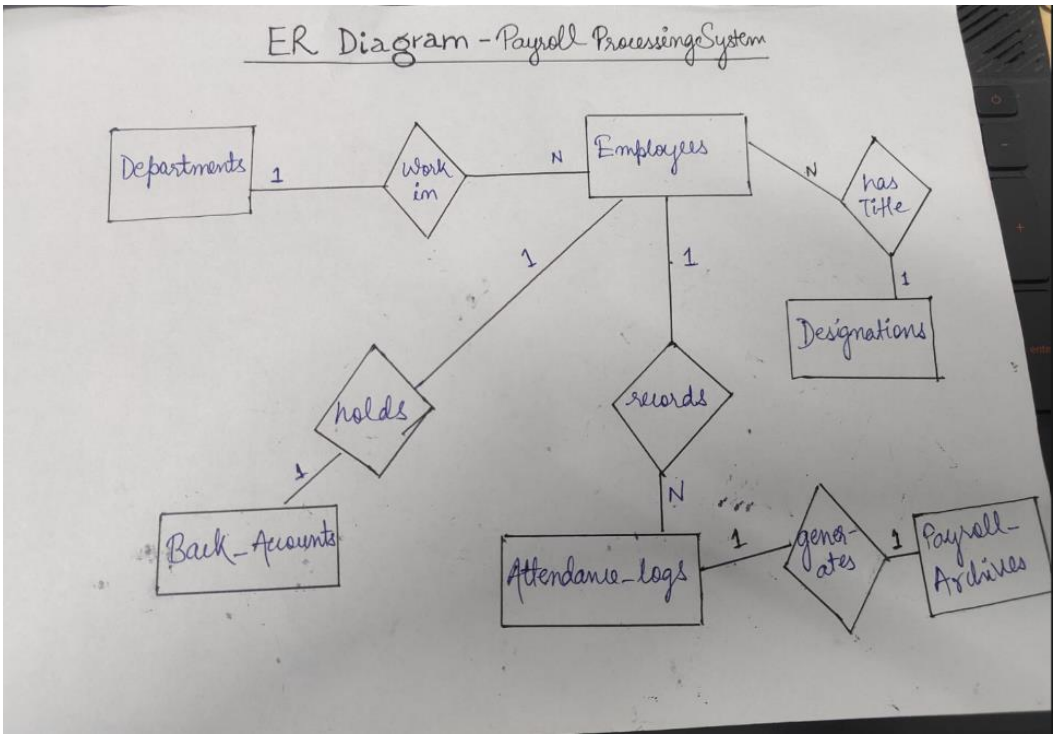
● Payroll Archives: Once monthly attendance is submitted, the system runs calculations to produce a unique payroll_tx_id. It records the gross_salary (base pay), subtracts a leave_penalty (for unpaid leaves), and saves the final net_salary.

● Bank Accounts: To pay employees, the system links the emp_id to a unique account_number, a bank_name, and an ifsc_code for electronic transfers.

Step 1: Entities and ER diagram

Each entity and its attributes (PK = primary key, FK = foreign key):

- **DEPARTMENTS**: dept_code (PK), dept_name, annual_budget
- **DESIGNATIONS**: desig_code (PK), job_title, base_salary
- **EMPLOYEES**: emp_id (PK), first_name, last_name, email, dept_code (FK), desig_code (FK)
- **BANK_ACCOUNTS**: account_number (PK), emp_id (FK), bank_name, ifsc_code
- **ATTENDANCE_LOGS**: log_id (PK), emp_id (FK), cal_month, cal_year, days_present, unpaid_leaves
- **PAYROLL_ARCHIVES**: payroll_tx_id (PK), log_id (FK), gross_salary, leave_penalty, net_salary



- A department has many employees, and each employee belongs to one department (1 to Many).
- A designation has many employees, and each employee has one designation (1 to Many).
- An employee has one bank account (1 to 1).
- An employee has many monthly attendance logs (1 to Many).
- Each attendance log produces one payroll record (1 to 1).

Step 2: Functional dependencies and BCNF

Table	Functional dependencies
departments	dept_code → dept_name, annual_budget
designations	desig_code → job_title, base_salary
employees	emp_id → first_name, last_name, email, dept_code, desig_code; email → emp_id
bank_accounts	account_number → emp_id, bank_name, ifsc_code; emp_id → account_number
attendance_logs	log_id → emp_id, cal_month, cal_year, days_present, unpaid_leaves; (emp_id, cal_month, cal_year) → log_id, days_present, unpaid_leaves
payroll_archives	payroll_tx_id → log_id, gross_salary, leave_penalty, net_salary; log_id → payroll_tx_id

BCNF check: a table is in BCNF if the left side of every functional dependency is a superkey.

- In every table above, each determinant is a primary key or a unique candidate key: email, emp_id in bank_accounts, the (emp_id, cal_month, cal_year) combination, and log_id in payroll_archives.
- So every table is in BCNF.
- Base salary is stored only in designations, not in employees. That avoids the dependency desig_code → base_salary inside the employees table, which would break BCNF.

Step 3: Table instances

The script inserts 10 rows into each of the six tables. To get the text tables for your report:

1. Run `mysql -u root -p < payroll_system.sql`.
2. Log in with `mysql -u root -p payroll_db`.
3. Run `SELECT * FROM departments;` and the same for the other five tables.
4. Screenshot or copy each output.

Step 4: DCL

The script creates the two users and gives them these rights:

- **hr_manager** gets `GRANT ALL PRIVILEGES`, so it can change records.
- **auditor** gets `GRANT SELECT` only.

To demonstrate it, log in with `mysql -u auditor -p payroll_db`. A `SELECT` works, and an `UPDATE` or `DELETE` fails with ERROR 1142. Screenshot that error for your report.

Step 5: SQL script

The file is organised into labelled sections:

- **DDL:** 6 tables with PK, FK, UNIQUE and CHECK constraints.
- **DML:** inserts, three updates and a delete. The payroll rows are computed by an `INSERT ... SELECT`, with penalty = base salary ÷ 30 × unpaid leaves.
- **DQL:** 5 queries, including a 6-table payslip join, a department summary with GROUP BY and a `LEFT JOIN`.
- **DCL:** users, grants and `SHOW GRANTS`.

SQL QUERY

```
CREATE DATABASE payroll_db;

USE payroll_db;


-- PART 1 : DDL (Data Definition Language)

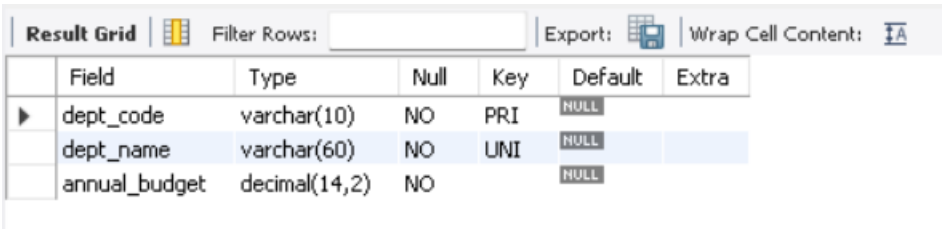
CREATE TABLE departments (

  dept_code    VARCHAR(10)  PRIMARY KEY,

  dept_name    VARCHAR(60)  NOT NULL UNIQUE,

  annual_budget DECIMAL(14,2) NOT NULL CHECK (annual_budget >= 0)

);
```



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
dept_code	varchar(10)	NO	PRI	NULL	
dept_name	varchar(60)	NO	UNI	NULL	
annual_budget	decimal(14,2)	NO		NULL	

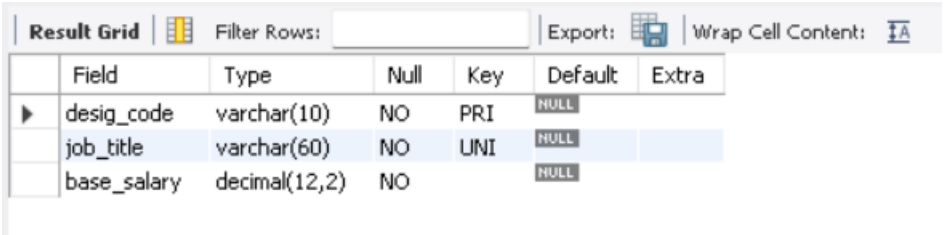
```
CREATE TABLE designations (

  desig_code  VARCHAR(10)  PRIMARY KEY,

  job_title   VARCHAR(60)  NOT NULL UNIQUE,

  base_salary DECIMAL(12,2) NOT NULL CHECK (base_salary > 0)

);
```



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
desig_code	varchar(10)	NO	PRI	NULL	
job_title	varchar(60)	NO	UNI	NULL	
base_salary	decimal(12,2)	NO		NULL	

```
CREATE TABLE employees (
    emp_id     INT         PRIMARY KEY,
    first_name VARCHAR(40)  NOT NULL,
    last_name   VARCHAR(40)  NOT NULL,
    email      VARCHAR(100) NOT NULL UNIQUE,
    dept_code  VARCHAR(10)  NOT NULL,
    desig_code VARCHAR(10)  NOT NULL,
    CONSTRAINT fk_emp_dept FOREIGN KEY (dept_code) REFERENCES departments(dept_code) ON UPDATE CASCADE ON DELETE RESTRICT,
    CONSTRAINT fk_emp_desig FOREIGN KEY (desig_code) REFERENCES designations(desig_code) ON UPDATE CASCADE ON DELETE RESTRICT
);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

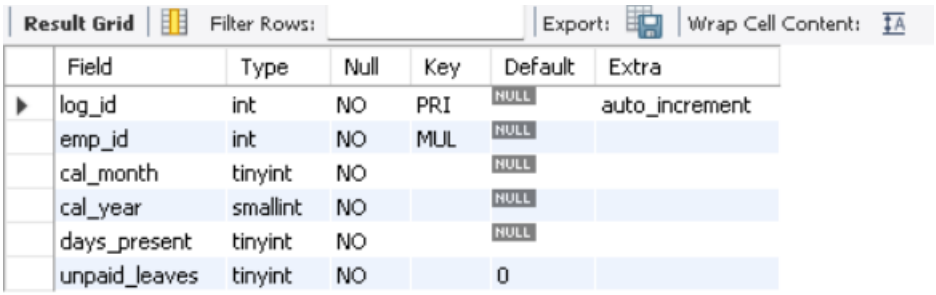
Field	Type	Null	Key	Default	Extra
emp_id	int	NO	PRI	NULL	
first_name	varchar(40)	NO		NULL	
last_name	varchar(40)	NO		NULL	
email	varchar(100)	NO	UNI	NULL	
dept_code	varchar(10)	NO	MUL	NULL	
desig_code	varchar(10)	NO	MUL	NULL	

```
CREATE TABLE bank_accounts (
    account_number VARCHAR(20) PRIMARY KEY,
    emp_id      INT         NOT NULL UNIQUE,      -- UNIQUE => 1-to-1 with employee
    bank_name     VARCHAR(50) NOT NULL,
    ifsc_code     CHAR(11)    NOT NULL,
   CONSTRAINT fk_bank_emp FOREIGN KEY (emp_id) REFERENCES employees(emp_id) ON UPDATE CASCADE ON DELETE CASCADE );
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
account_number	varchar(20)	NO	PRI	NULL	
emp_id	int	NO	UNI	NULL	
bank_name	varchar(50)	NO		NULL	
ifsc_code	char(11)	NO		NULL	

```
CREATE TABLE attendance_logs (
   log_id      INT AUTO_INCREMENT PRIMARY KEY,
   emp_id      INT      NOT NULL,
   cal_month    TINYINT   NOT NULL CHECK (cal_month BETWEEN 1 AND 12),
   cal_year     SMALLINT NOT NULL CHECK (cal_year >= 2000),
   days_present TINYINT   NOT NULL CHECK (days_present BETWEEN 0 AND 31),
   unpaid_leaves TINYINT  NOT NULL DEFAULT 0 CHECK (unpaid_leaves BETWEEN 0 AND 31),
   CONSTRAINT uq_emp_month UNIQUE (emp_id, cal_month, cal_year),   -- one log per employee per month
   CONSTRAINT fk_att_emp FOREIGN KEY (emp_id) REFERENCES employees(emp_id) ON UPDATE CASCADE ON
DELETE RESTRICT);
```



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
log_id	int	NO	PRI	NULL	auto_increment
emp_id	int	NO	MUL	NULL	
cal_month	tinyint	NO		NULL	
cal_year	smallint	NO		NULL	
days_present	tinyint	NO		NULL	
unpaid_leaves	tinyint	NO		0	

```
CREATE TABLE payroll_archives (
   payroll_tx_id INT AUTO_INCREMENT PRIMARY KEY,
   log_id       INT          NOT NULL UNIQUE,          -- one payroll record per attendance log
   gross_salary  DECIMAL(12,2) NOT NULL,
   leave_penalty DECIMAL(12,2) NOT NULL DEFAULT 0,
   net_salary    DECIMAL(12,2) NOT NULL,
   CONSTRAINT chk_net CHECK (net_salary = gross_salary - leave_penalty),
   CONSTRAINT fk_pay_log FOREIGN KEY (log_id) REFERENCES attendance_logs(log_id) ON
UPDATE CASCADE ON DELETE RESTRICT
);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
payroll_tx_id	int	NO	PRI	NULL	auto_increment
log_id	int	NO	UNI	NULL	
gross_salary	decimal(12,2)	NO		NULL	
leave_penalty	decimal(12,2)	NO		0.00	
net_salary	decimal(12,2)	NO		NULL	

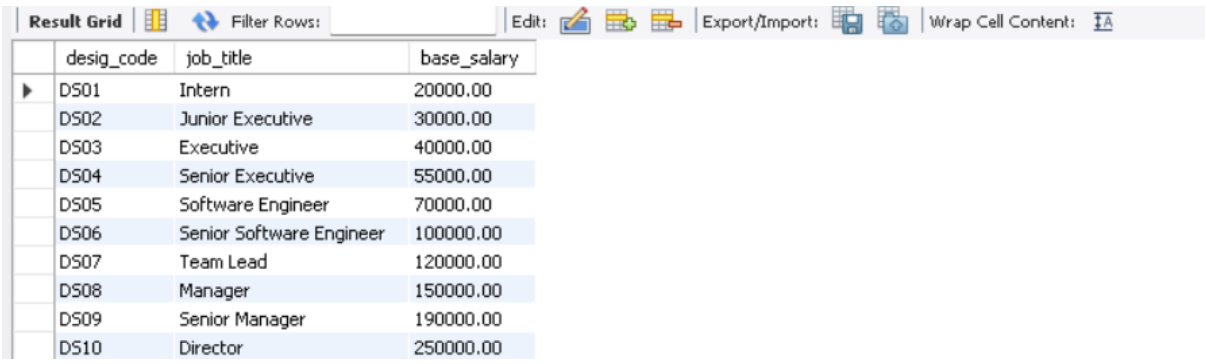
-- PART 2 : DML (Data Manipulation Language) - INSERT

```
INSERT INTO departments VALUES
('D01','Human Resources',    5000000.00),
('D02','Finance',            8000000.00),
('D03','Engineering',       25000000.00),
('D04','Sales',             12000000.00),
('D05','Marketing',          9000000.00),
('D06','Operations',        11000000.00),
('D07','IT Support',         6000000.00),
('D08','Legal',              4000000.00),
('D09','Research',          15000000.00),
('D10','Customer Support',   7000000.00);
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

dept_code	dept_name	annual_budget
D01	Human Resources	5000000.00
D02	Finance	8000000.00
D03	Engineering	25000000.00
D04	Sales	12000000.00
D05	Marketing	9000000.00
D06	Operations	11000000.00
D07	IT Support	6000000.00
D08	Legal	4000000.00
D09	Research	15000000.00
D10	Customer Support	7000000.00

```
INSERT INTO designations VALUES
('DS01','Intern',                20000.00),
('DS02','Junior Executive',      30000.00),
('DS03','Executive',             40000.00),
('DS04','Senior Executive',      55000.00),
('DS05','Software Engineer',     70000.00),
('DS06','Senior Software Engineer',100000.00),
('DS07','Team Lead',             120000.00),
('DS08','Manager',               150000.00),
('DS09','Senior Manager',        190000.00),
('DS10','Director',              250000.00);
```



Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

desig_code	job_title	base_salary
DS01	Intern	20000.00
DS02	Junior Executive	30000.00
DS03	Executive	40000.00
DS04	Senior Executive	55000.00
DS05	Software Engineer	70000.00
DS06	Senior Software Engineer	100000.00
DS07	Team Lead	120000.00
DS08	Manager	150000.00
DS09	Senior Manager	190000.00
DS10	Director	250000.00

```
INSERT INTO employees VALUES
(101,'Ayush', 'Kumar','ayush.kumar@company.com','D03','DS05'),
(102,'Abhijeet', 'Lakra', 'abhijeet.lakra@company.com', 'D01','DS08'),
(103,'Nikhil', 'Raj', 'nikhil.raj@company.com', 'D02','DS04'),
(104,'Harshit', 'Kumar', 'harshit.kumar@company.com', 'D03','DS06'),
(105,'Mahesh','Babu', 'mahesh.babu@company.com','D04','DS07'),
```

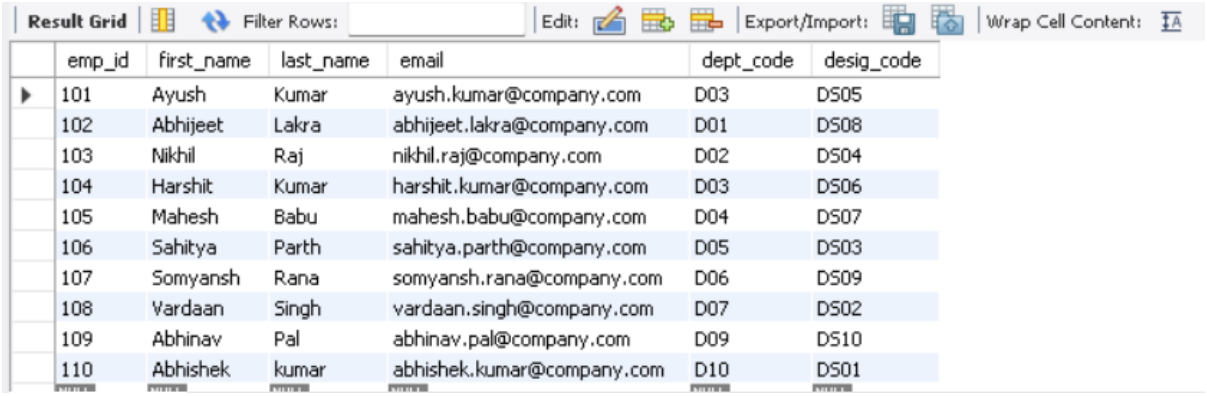
(106,'Sahitya','Parth', 'sahitya.parth@company.com', 'D05','DS03'),

(107,'Somyansh', 'Rana', 'somyansh.rana@company.com', 'D06','DS09'),

(108,'Vardaan', 'Singh','vardaan.singh@company.com', 'D07','DS02'),

(109,'Abhinav', 'Pal', 'abhinav.pal@company.com', 'D09','DS10'),

(110,'Abhishek', 'kumar', 'abhishek.kumar@company.com', 'D10','DS01');



emp_id	first_name	last_name	email	dept_code	desig_code
101	Ayush	Kumar	ayush.kumar@company.com	D03	DS05
102	Abhijeet	Lakra	abhijeet.lakra@company.com	D01	DS08
103	Nikhil	Raj	nikhil.raj@company.com	D02	DS04
104	Harshit	Kumar	harshit.kumar@company.com	D03	DS06
105	Mahesh	Babu	mahesh.babu@company.com	D04	DS07
106	Sahitya	Parth	sahitya.parth@company.com	D05	DS03
107	Somyansh	Rana	somyansh.rana@company.com	D06	DS09
108	Vardaan	Singh	vardaan.singh@company.com	D07	DS02
109	Abhinav	Pal	abhinav.pal@company.com	D09	DS10
110	Abhishek	kumar	abhishek.kumar@company.com	D10	DS01

INSERT INTO bank_accounts VALUES

('100200300401',101,'State Bank of India','SBIN0001234'),

('100200300402',102,'HDFC Bank', 'HDFC0000456'),

('100200300403',103,'ICICI Bank', 'ICIC0000789'),

('100200300404',104,'Axis Bank', 'UTIB0000321'),

('100200300405',105,'Punjab National Bank','PUNB0123400'),

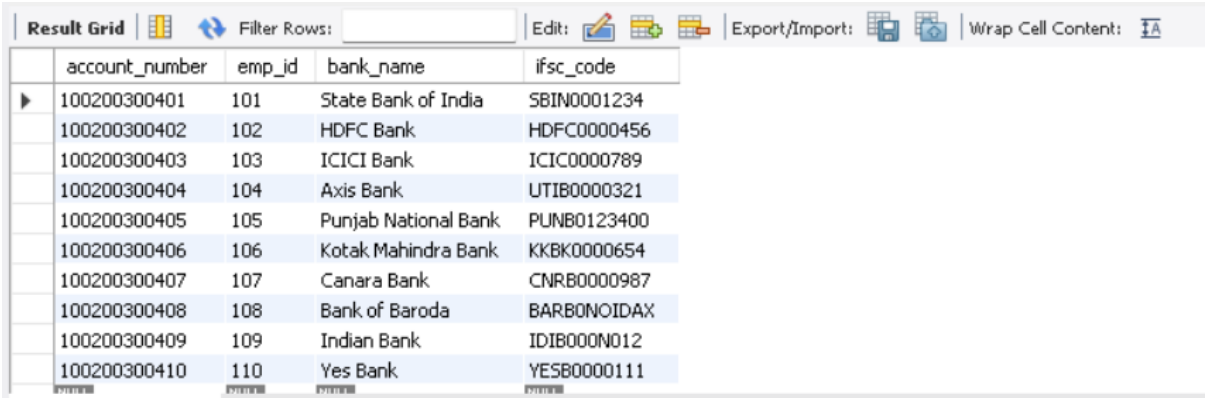
('100200300406',106,'Kotak Mahindra Bank','KKBK0000654'),

('100200300407',107,'Canara Bank', 'CNRB0000987'),

('100200300408',108,'Bank of Baroda', 'BARB0NOIDAX'),

('100200300409',109,'Indian Bank', 'IDIB000N012'),

('100200300410',110,'Yes Bank', 'YESB0000111');



account_number	emp_id	bank_name	ifsc_code
100200300401	101	State Bank of India	SBIN0001234
100200300402	102	HDFC Bank	HDFC0000456
100200300403	103	ICICI Bank	ICIC0000789
100200300404	104	Axis Bank	UTIB0000321
100200300405	105	Punjab National Bank	PUNB0123400
100200300406	106	Kotak Mahindra Bank	KKBK0000654
100200300407	107	Canara Bank	CNRB0000987
100200300408	108	Bank of Baroda	BARB0NOIDAX
100200300409	109	Indian Bank	IDIB000N012
100200300410	110	Yes Bank	YESB0000111

-- Attendance for September 2026

INSERT INTO attendance_logs (emp_id, cal_month, cal_year, days_present, unpaid_leaves) VALUES

(101,9,2026,28,2),

(102,9,2026,30,0),

(103,9,2026,29,1),

(104,9,2026,27,3),

(105,9,2026,30,0),

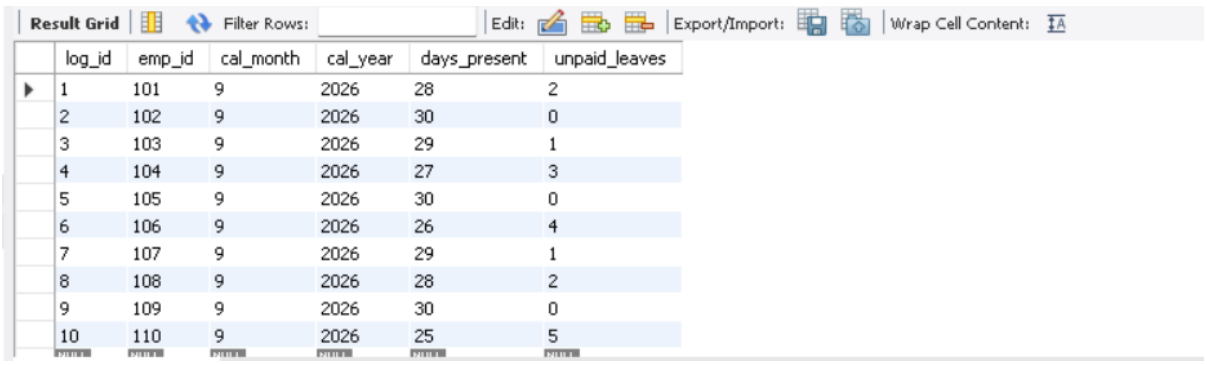
(106,9,2026,26,4),

(107,9,2026,29,1),

(108,9,2026,28,2),

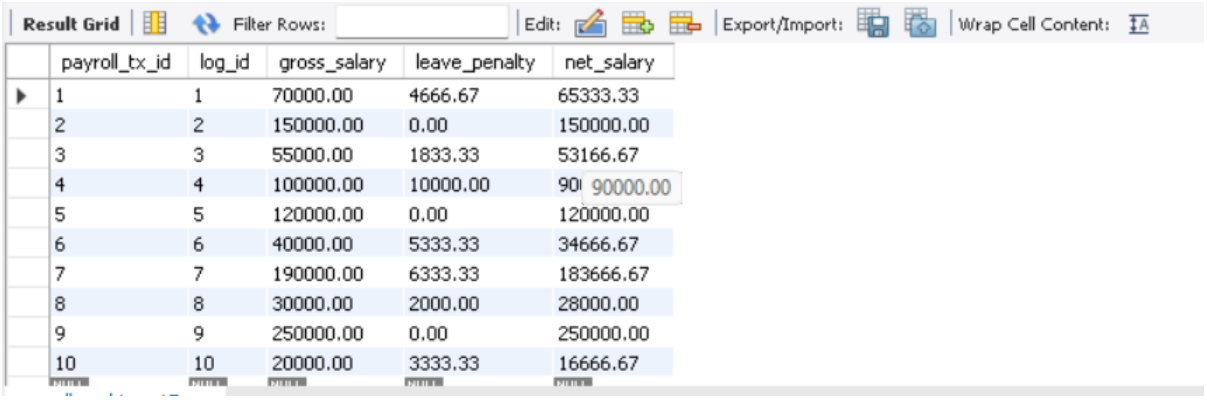
(109,9,2026,30,0),

(110,9,2026,25,5);



log_id	emp_id	cal_month	cal_year	days_present	unpaid_leaves
1	101	9	2026	28	2
2	102	9	2026	30	0
3	103	9	2026	29	1
4	104	9	2026	27	3
5	105	9	2026	30	0
6	106	9	2026	26	4
7	107	9	2026	29	1
8	108	9	2026	28	2
9	109	9	2026	30	0
10	110	9	2026	25	5

```
-- Payroll run: gross = base salary, penalty = (base / 30) * unpaid leaves
INSERT INTO payroll_archives (log_id, gross_salary, leave_penalty, net_salary)
SELECT a.log_id,
       d.base_salary,
       ROUND(d.base_salary / 30 * a.unpaid_leaves, 2),
       d.base_salary - ROUND(d.base_salary / 30 * a.unpaid_leaves, 2)
FROM   attendance_logs a
JOIN   employees    e ON e.emp_id    = a.emp_id
JOIN   designations d ON d.desig_code = e.desig_code
WHERE  a.cal_month = 9 AND a.cal_year = 2026;
```



Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

payroll_tx_id	log_id	gross_salary	leave_penalty	net_salary
1	1	70000.00	4666.67	65333.33
2	2	150000.00	0.00	150000.00
3	3	55000.00	1833.33	53166.67
4	4	100000.00	10000.00	90 90000.00
5	5	120000.00	0.00	120000.00
6	6	40000.00	5333.33	34666.67
7	7	190000.00	6333.33	183666.67
8	8	30000.00	2000.00	28000.00
9	9	250000.00	0.00	250000.00
10	10	20000.00	3333.33	16666.67

```
--  PART 3 : DQL  (Data Query Language) - multi-table joins
 -- Q1: Full payslip for every employee (6-table join)
SELECT p.payroll_tx_id,
       CONCAT(e.first_name,' ',e.last_name) AS employee,
       dp.dept_name, ds.job_title,
       a.cal_month, a.cal_year, a.unpaid_leaves,
```

```
    p.gross_salary, p.leave_penalty, p.net_salary,
    b.bank_name, b.account_number, b.ifsc_code
FROM   payroll_archives p
JOIN   attendance_logs a ON a.log_id    = p.log_id
JOIN   employees       e ON e.emp_id    = a.emp_id
JOIN   departments     dp ON dp.dept_code = e.dept_code
JOIN   designations    ds ON ds.desig_code = e.desig_code
JOIN   bank_accounts   b ON b.emp_id    = e.emp_id
ORDER  BY p.payroll_tx_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

pay	employee	dept_name	job_title	cal	cal_y	unp	gross_salar	leave_pen	net_salary	bank_name	account_number	ifsc_code
1	Ayush Kumar	Engineering	Software Engineer	9	2026	2	70000.00	4666.67	65333.33	State Bank of India	100200300401	SBIN0001234
2	Abhijeet L...	Human Re...	Manager	9	2026	0	150000.00	0.00	150000.00	HDFC Bank	100200300402	HDFC0000456
3	Nikhil Raj	Finance	Senior Executive	9	2026	1	55000.00	1833.33	53166.67	ICICI Bank	100200300403	ICIC0000789
4	Harshit Ku...	Engineering	Senior Software ...	9	2026	3	100000.00	10000.00	90000.00	Axis Bank	100200300404	UTIB0000321
5	Mahesh Babu	Sales	Team Lead	9	2026	0	120000.00	0.00	120000.00	Punjab National Bank	100200300405	PUNB0123400
6	Sahitya Parth	Marketing	Executive	9	2026	4	40000.00	5333.33	34666.67	Kotak Mahindra Bank	100200300406	KKBK0000654
7	Somyansh ...	Operations	Senior Manager	9	2026	1	190000.00	6333.33	183666.67	Canara Bank	100200300407	CNRB0000987
8	Vardaan Si...	IT Support	Junior Executive	9	2026	2	30000.00	2000.00	28000.00	Bank of Baroda	100200300408	BARB0NOIDAX
9	Abhinav Pal	Research	Director	9	2026	0	250000.00	0.00	250000.00	Indian Bank	100200300409	IDIB000N012
10	Abhishek k...	Customer ...	Intern	9	2026	5	20000.00	3333.33	16666.67	Yes Bank	100200300410	YESB0000111

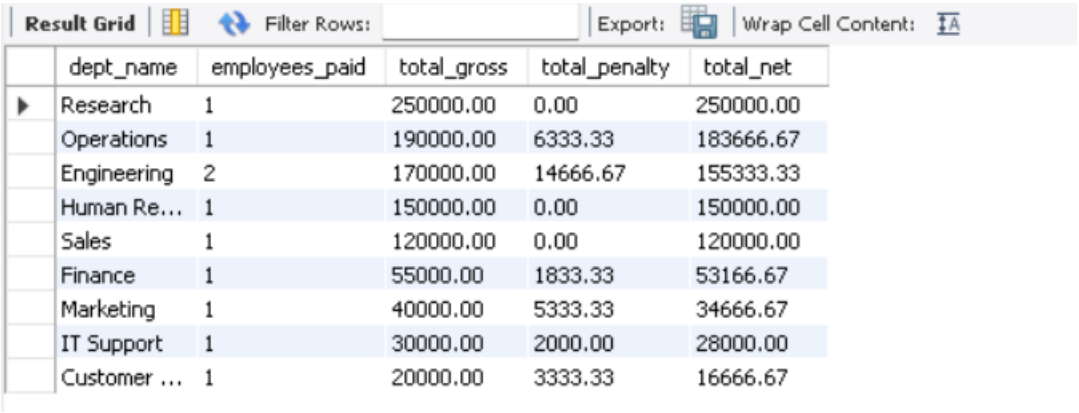
```
-- Q2: Department-wise payout summary
SELECT dp.dept_name,
    COUNT(*)           AS employees_paid,
    SUM(p.gross_salary) AS total_gross,
    SUM(p.leave_penalty) AS total_penalty,
    SUM(p.net_salary)   AS total_net
FROM   payroll_archives p
JOIN   attendance_logs a ON a.log_id = p.log_id
```

```
JOIN   employees e       ON e.emp_id = a.emp_id

JOIN   departments dp    ON dp.dept_code = e.dept_code

GROUP  BY dp.dept_name

ORDER  BY total_net DESC;
```



dept_name	employees_paid	total_gross	total_penalty	total_net
Research	1	250000.00	0.00	250000.00
Operations	1	190000.00	6333.33	183666.67
Engineering	2	170000.00	14666.67	155333.33
Human Re...	1	150000.00	0.00	150000.00
Sales	1	120000.00	0.00	120000.00
Finance	1	55000.00	1833.33	53166.67
Marketing	1	40000.00	5333.33	34666.67
IT Support	1	30000.00	2000.00	28000.00
Customer ...	1	20000.00	3333.33	16666.67

```
-- Q3: Employees who took more than 2 unpaid leaves

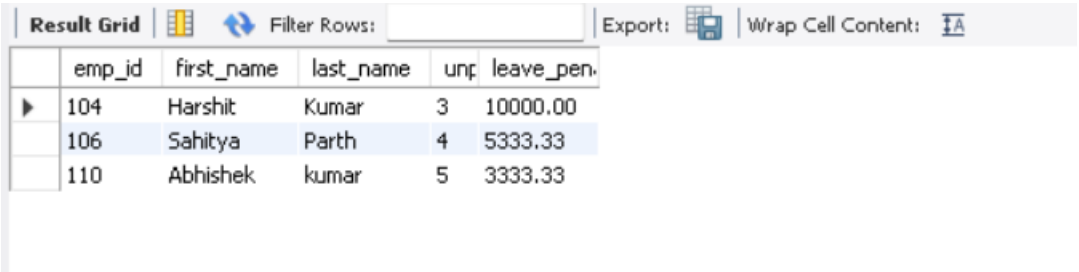
SELECT e.emp_id, e.first_name, e.last_name, a.unpaid_leaves, p.leave_penalty

FROM   employees e

JOIN   attendance_logs a  ON a.emp_id = e.emp_id

JOIN   payroll_archives p ON p.log_id = a.log_id

WHERE  a.unpaid_leaves > 2;
```



emp_id	first_name	last_name	unp	leave_pen.
104	Harshit	Kumar	3	10000.00
106	Sahitya	Parth	4	5333.33
110	Abhishek	kumar	5	3333.33

```
-- Q4: Designation-wise average net salary
```

```
SELECT ds.job_title, ROUND(AVG(p.net_salary),2) AS avg_net

FROM   payroll_archives p

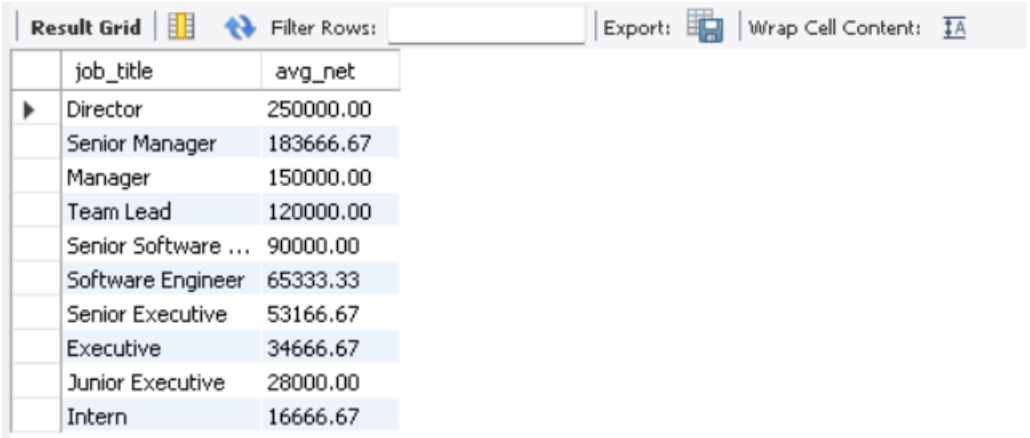
JOIN   attendance_logs a ON a.log_id = p.log_id

JOIN   employees e       ON e.emp_id = a.emp_id

JOIN   designations ds   ON ds.desig_code = e.desig_code

GROUP  BY ds.job_title

ORDER  BY avg_net DESC;
```



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

job_title	avg_net
Director	250000.00
Senior Manager	183666.67
Manager	150000.00
Team Lead	120000.00
Senior Software ...	90000.00
Software Engineer	65333.33
Senior Executive	53166.67
Executive	34666.67
Junior Executive	28000.00
Intern	16666.67

```
-- Q5: Departments with no employees (LEFT JOIN)

SELECT dp.dept_code, dp.dept_name

FROM   departments dp

LEFT JOIN employees e ON e.dept_code = dp.dept_code

WHERE  e.emp_id IS NULL;
```



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

dept_code	dept_name
D08	Legal

```
-- PART 4 : DCL (Data Control Language)
```

```
DROP USER IF EXISTS 'hr_manager'@'localhost';

DROP USER IF EXISTS 'auditor'@'localhost';

CREATE USER 'hr_manager'@'localhost' IDENTIFIED BY 'hr123';

CREATE USER 'auditor'@'localhost' IDENTIFIED BY 'audit123';

GRANT SELECT, INSERT, UPDATE, DELETE ON payroll_db.* TO 'hr_manager'@'localhost';

GRANT SELECT ON payroll_db.* TO 'auditor'@'localhost';

SHOW GRANTS FOR 'hr_manager'@'localhost';

SHOW GRANTS FOR 'auditor'@'localhost';
```

